DATA WRANGLING GROUP PROJECT

Breathe India

What really drives air pollution in Indian cities?

26
cities

5½ years
of daily data, 2015 to 2020

4
sources joined

3
machine-learning models

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THE ANSWER, UP FRONT

A third of the lockdown's “clean air” was actually the weather

−41%

PM2.5 change in spring 2020 vs 2019: the headline everyone quoted

−28%

the real lockdown effect once our model removes the weather

75%

of next-day air-quality categories predicted exactly (baseline 72%)

Also: cleaning changed the story. Officially, Ahmedabad is India's most polluted city. It isn't: a sensor column was copied.

DATA AND PIPELINE

Four sources, one star schema, three storage formats

CPCB air quality
Kaggle download · 29,531 city-days, 108,035 station-days

Open-Meteo weather API
52,234 city-days of ERA5 weather · rate-limited, cached

Open-Meteo geocoding API
26 cities located · names checked by state

Event calendar
Lockdown phases and Diwali dates

Clean fact table
28,260 city-days × 61 columns

23 logged cleaning decisions

Validated joins, lags, rolling means

SQLite
keys + read-back checks

Parquet
partitioned by year

Power BI
5-page dashboard

Models A, B, C
scikit-learn

Everything reruns end to end in about 8 minutes from one notebook: github.com/mahak2604/breathe-india

CLEANING

The raw data was wrong in ways a quick look would miss

1 A copied column

Ahmedabad's CO equals its NO on 94% of days, although the units differ 1000×. It drove 95% of the city's “Severe” days.

2 Impossible pairs

PM2.5 above PM10 on 256 days. We assumed PM10 was wrong; the data showed PM2.5 was the faulty one 94% of the time.

3 Broken sensors

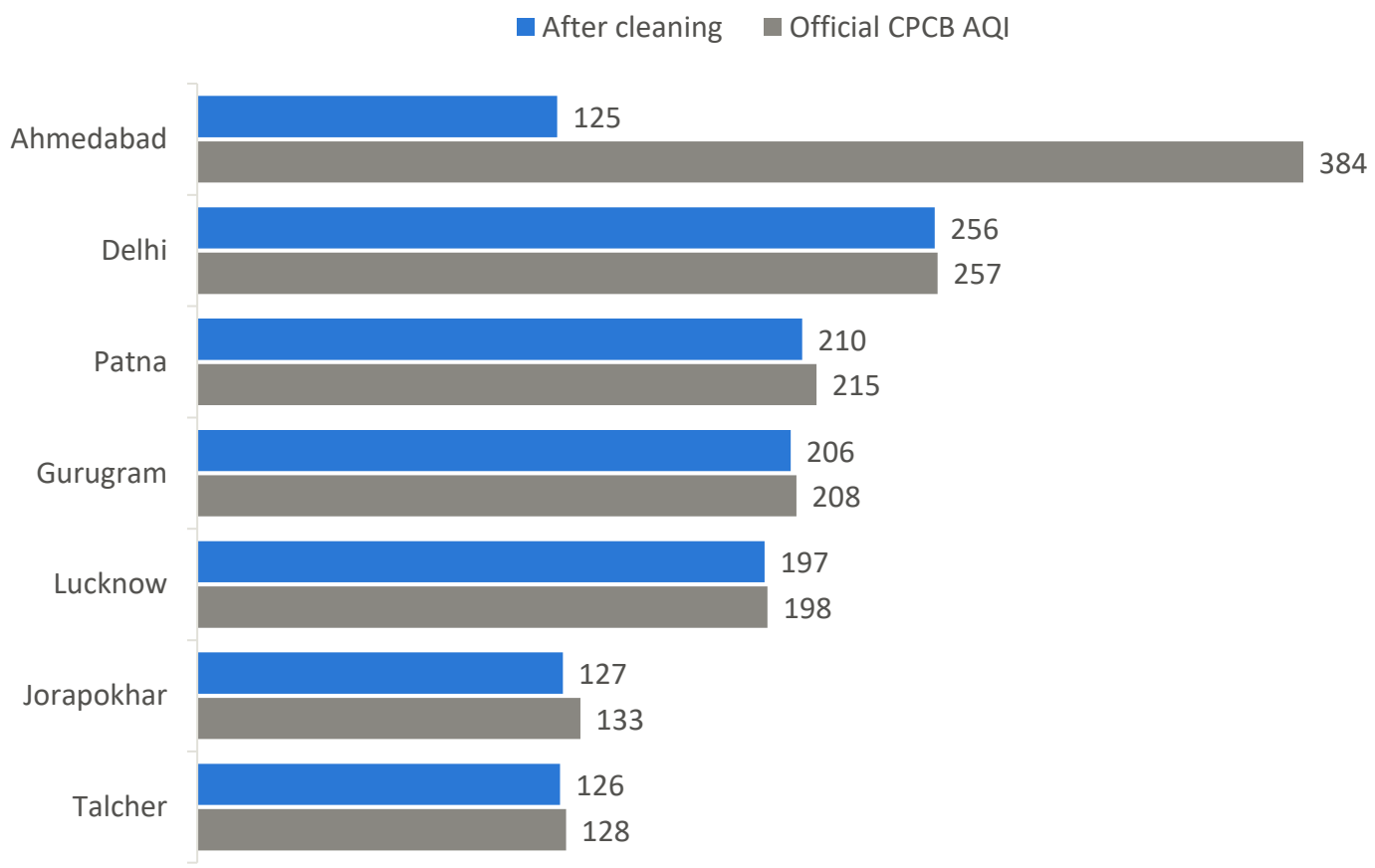
A sensor stuck at 11.00 for 16 days, and 918 µg/m³ in Shillong, the cleanest city. 197 readings removed, with Diwali spikes kept.

4 Gaps and limits

AQI up to 2,049 on a 0–500 scale, exact zeros, 61% of one column empty. Short gaps interpolated, the rest left honest.

INSIGHT 1

Cleaning moved Ahmedabad from India's most polluted city to 7th



Median daily AQI, 2015–2020, top 7 cities on official numbers

384 → 125

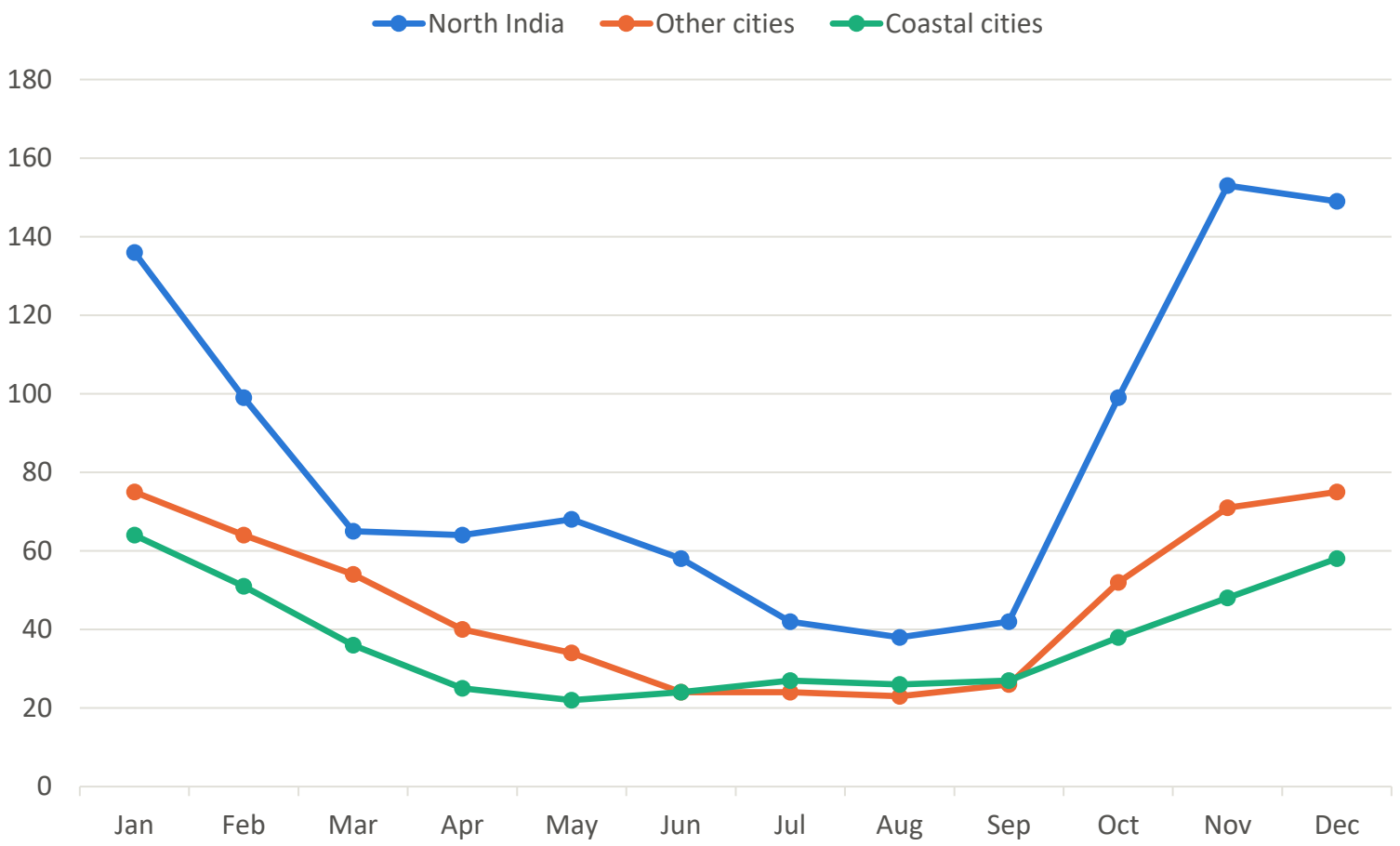
Ahmedabad's median AQI, official vs cleaned

Every other city barely moves.

Without this one fix, any ranking, map or model would have named the wrong city.

INSIGHTS 2 AND 3

North India's winter air carries 3.3× the PM2.5 of its monsoon air



Median PM2.5 (µg/m³) by month, 2015–2020 · India's 24-hour limit is 60

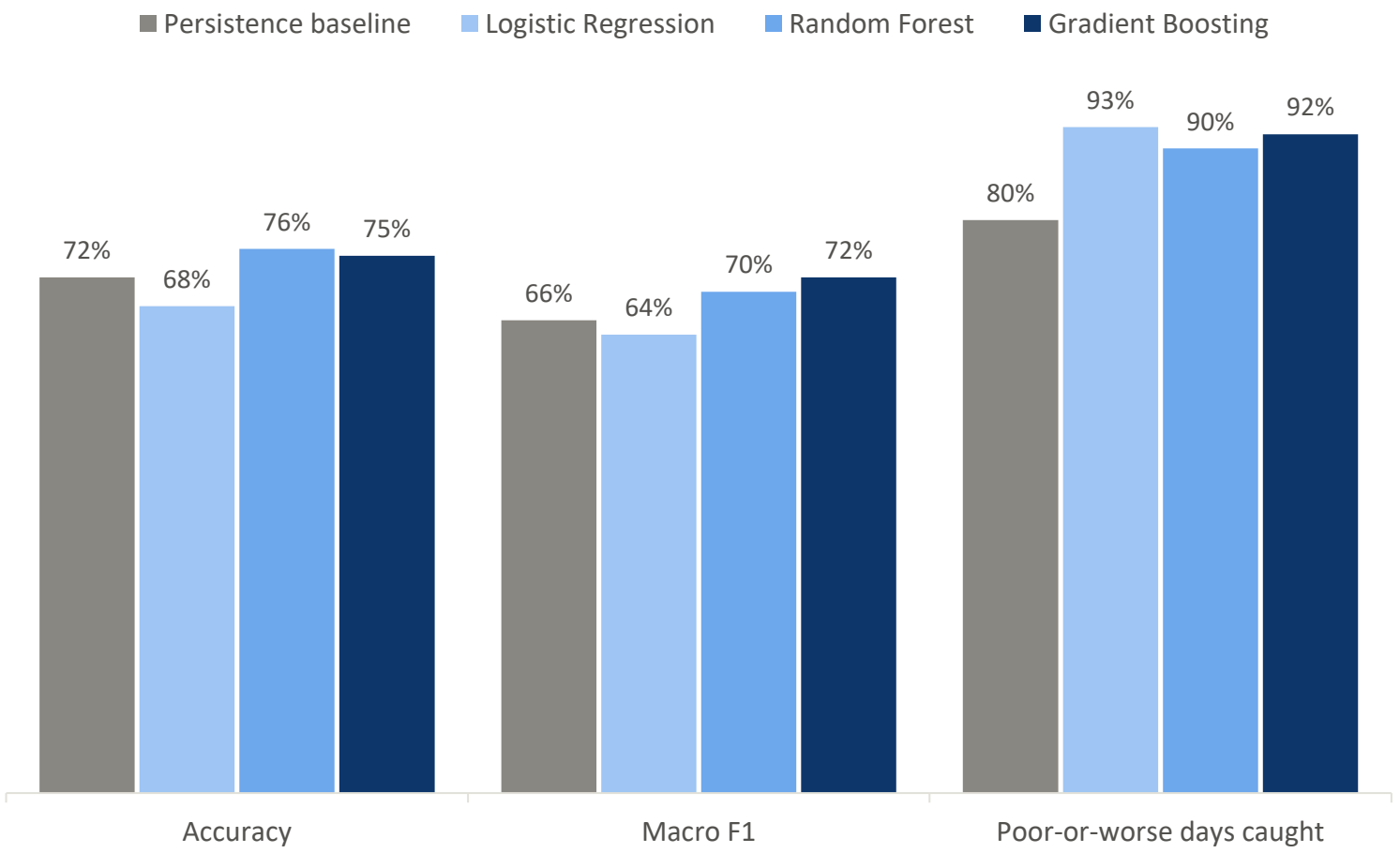
148 vs 44

µg/m³: North India in Nov–Jan vs Jun–Sep. Coastal cities swing only 2.2×.

23%, not 65%

how much cleaner a heavy-rain day really is, once we compare the same city in the same month

Model A warns a day ahead and catches 92% of Poor-or-worse days

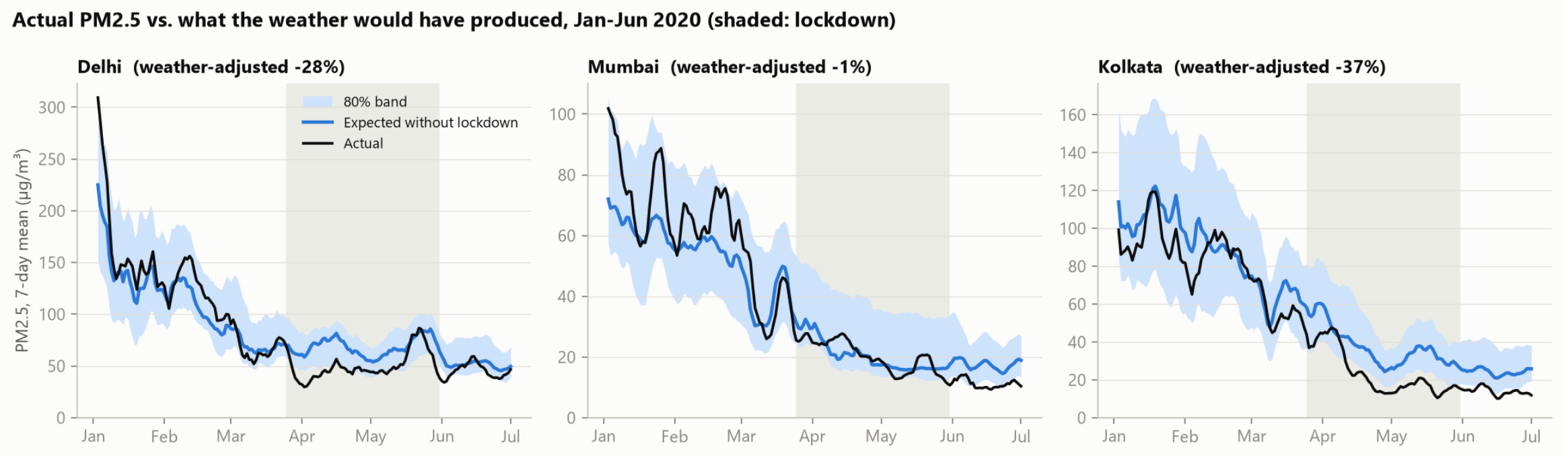


Gradient Boosting beats "same as today" on accuracy (75% vs 72%) and is right 35% of the time when the category changes, where the baseline is always wrong.

Honest caveat: in the lockdown its macro F1 fell to 0.59, below the baseline's 0.65. The rules it learned stopped applying.

Unseen test period Jan 2019 – Mar 2020 · time-ordered split, time-series cross-validation

Weather removed, the lockdown cut PM2.5 by 28%, not 41%



–41% → –28%

PM2.5: naive vs weather-adjusted, median of 17 cities

–47% → –46%

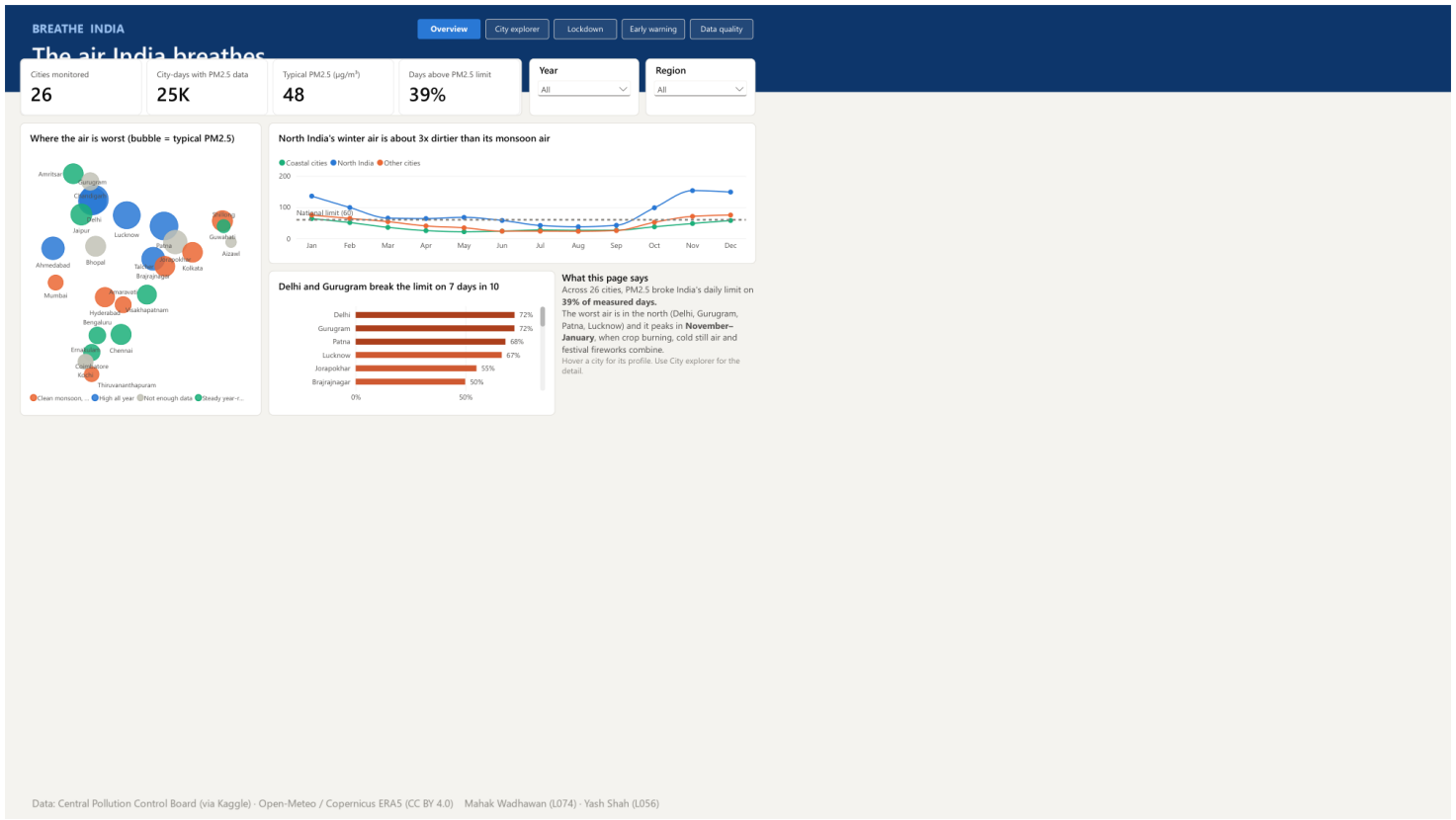
NO_2 (mostly traffic): a real drop either way

R^2 0.55

placebo test on spring 2019, when nothing happened

POWER BI

The dashboard turns this into five questions a non-expert can explore



1. **Overview**
   where and when the air is worst
2. **City explorer**
   one city's year, stations, AQI bands
3. **Lockdown**
   actual vs expected, pick any city
4. **Early warning**
   model vs baseline, switch periods
5. **Data quality**
   every cleaning decision and why

Live demo: Breathe_India.pbix · hover a city, drill into Delhi, switch Test → Lockdown

What we can say, and what we can't

Limitations

Monitors cover parts of 26 cities, not everyone's exposure

Our AQI corrections are recalculations, not CPCB's own

Model A uses actual weather as a forecast stand-in

Model B assumes 2015–19 “normal” would have held in 2020

Data ends 1 July 2020

Takeaways

Season sets the baseline: winter is the problem in the north

People still matter: −28% PM2.5 and −46% NO_2 without traffic

Wrangling changed the answer: check the data before trusting a ranking

Contribution: Mahak led transformation, the models and the dashboard; Yash led extraction, cleaning and storage; insights, report and deck together.

Thank you. Questions?